# Bird Species Classification with Convolutional Neural Networks

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**Abstract**

This report evaluates a custom convolutional neural network (CNN A) and two transfer-learning variants (CNN B) for classifying bird species. CNN A, was created through staged hyperparameter tuning, achieving a 98% test accuracy on Part 1. In Part 2, transfer learning of CNN A achieved 68% test accuracy by training a new classification head, compared with 52% when the final convolutional block was fine-tuned. Reusing a frozen feature extractor reduced the number of parameters updated.

## 1 Aim and Experimental Design

This report investigates whether a CNN optimised for one group of bird species can provide useful visual features for a second group. Each split contains 50 validation and 50 test images; the training splits contain 1,546 images in Part 1 and 1,561 in Part 2. The test images are used for final evaluation, while the validation images are used for early stopping and hyperparameter tuning.

## 2 Methods

### 2.1 CNN A: Architecture Selection

To create CNN A from scratch, candidate CNNs were screened using reduced-resolution 128×128 images, using Keras hyperparameter tuning. The top three models were retrained at 224×224 resolution. Because the initial search indicated strong performance from a four-block configuration, a focussed search of four-block architectures was conducted separately, using a longer training schedule to assess them more fully. The search varied network depth, filter count, kernel size, dense-layer width, and learning rate. CNN A was selected by validation performance from the full-resolution comparison.

The selected model has four convolutional blocks with 64, 128, 128, and 32 filters, respectively. Each block applies convolution, batch normalisation, and max pooling; global average pooling and a 64-unit dense layer precede dropout and a 10-class softmax output. This architecture was selected because it achieved the strongest observed full-resolution test performance while maintaining a compact classifier head. Other four-block candidates were also competitive, but did not exceed CNN A in the recorded comparison.

### 2.2 CNN B: Transfer-Learning Variants

To build CNN B, the convolutional feature extractor from CNN A was loaded for transfer learning for the classes of bird species found in Part 2. Two variations of tranfer learning were attempted in the notebook, retraining the classification head only, or retraining the head and the final convolutional block. Both were trained using the same test, train and validation splits of the full resolution 224×224 images. The head-only configuration was selected as CNN B as it achieved the strongest validation performance.

### 2.3 Preprocessing, Augmentation, and Measures

Images were rescaled to $[0, 1]$. Training images were augmented using rotations, horizontal and vertical shifts, shear, zoom, and horizontal reflection. Augmentation was applied to the full-resolution training data to expose the models to pose and viewpoint variation and subsequently improve generalisation. Inverse-frequency class weights, $w_c = N/(Kn_c)$, were also used to account for class imbalance, where $N$ is the

| Layer (type) | Output Shape | Param # |
|---|---|---|
| input_layer_4 (InputLayer) | (None, 224, 224, 3) | 0 |
| conv2d_13 (Conv2D) | (None, 224, 224, 64) | 1,792 |
| batch_normalization_13 (BatchNormalization) | (None, 224, 224, 64) | 256 |
| max_pooling2d_13 (MaxPooling2D) | (None, 112, 112, 64) | 0 |
| conv2d_14 (Conv2D) | (None, 112, 112, 128) | 204,928 |
| batch_normalization_14 (BatchNormalization) | (None, 112, 112, 128) | 512 |
| max_pooling2d_14 (MaxPooling2D) | (None, 56, 56, 128) | 0 |
| conv2d_15 (Conv2D) | (None, 56, 56, 128) | 147,584 |
| batch_normalization_15 (BatchNormalization) | (None, 56, 56, 128) | 512 |
| max_pooling2d_15 (MaxPooling2D) | (None, 28, 28, 128) | 0 |
| conv2d_16 (Conv2D) | (None, 28, 28, 32) | 36,896 |
| batch_normalization_16 (BatchNormalization) | (None, 28, 28, 32) | 128 |
| max_pooling2d_16 (MaxPooling2D) | (None, 14, 14, 32) | 0 |
| global_average_pooling2d_4 (GlobalAveragePooling2D) | (None, 32) | 0 |
| dense_2 (Dense) | (None, 256) | 8,448 |
| dropout_1 (Dropout) | (None, 256) | 0 |
| dense_3 (Dense) | (None, 10) | 2,570 |

Figure 1: Terminal-generated layer summary of CNN A. The model has four convolutional blocks, followed by global average pooling, a dense layer, dropout, and a 10-class softmax output.

training-set size, $K$ is the number of classes, and $n_c$ is the number of training images in class $c$. Validation loss was monitored for early stopping and checkpoint restoration.

In this report, performance is summarised using accuracy, precision, recall, and F1 score, alongside test loss. The reported best epoch is the epoch with maximum validation accuracy. Early stopping and weight restoration are based separately on minimum validation loss.

## 3 Results

Table 1: Comparison of the evaluated Part 1 candidates and Part 2 transfer-learning variants. Part 1 and Part 2 results refer to different bird species sets.

| Model | Val. acc. | Test acc. | Macro F1 | Test loss | Epoch[a] |
|---|---|---|---|---|---|
| *Part 1: CNN A Candidates* | | | | | |
| Initial-search candidate 1 (CNN A) | 0.96 | 0.98 | 0.980 | 0.097 | 49 |
| Initial-search candidate 2 | 0.88 | 0.80 | 0.781 | 0.578 | 47 |
| Initial-search candidate 3 | 0.84 | 0.70 | 0.664 | 0.884 | 40 |
| Four-block candidate 1 | 0.96 | 0.94 | 0.938 | 0.293 | 29 |
| Four-block candidate 2 | 0.96 | 0.96 | 0.960 | 0.123 | 33 |
| Four-block candidate 3 | 0.90 | 0.84 | 0.838 | 0.539 | 27 |
| *Part 2: CNN B Candidates* | | | | | |
| Head-only transfer (CNN B) | 0.64 | 0.68 | 0.669 | 1.275 | 16 |
| Fine-tuned final block | 0.54 | 0.52 | 0.48 | 1.662 | 76 |

### 3.1 Part 1: CNN A Candidate Comparison

CNN A's full-resolution candidate 1 achieved 0.96 validation accuracy and 0.98 test accuracy (macro F1 = 0.980). The strongest four-block candidate reached 0.96 test accuracy. Several alternative candidates reached peak validation accuracy in fewer epochs. CNN A attained its peak validation accuracy at epoch 49, and was selected for predictive performance rather than the earliest peak among candidates.

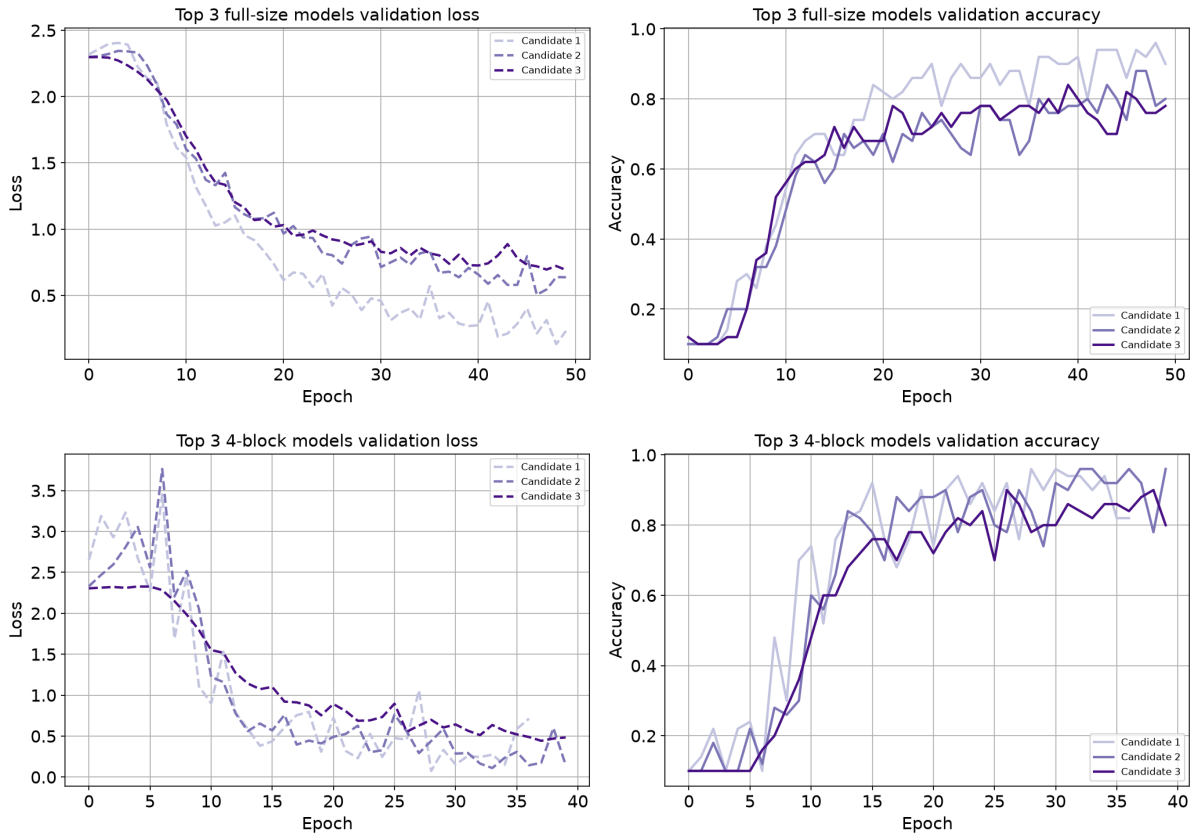



Figure 2: Validation accuracy and loss for the full-resolution candidate search (top) and the focused four-block search (bottom).

### 3.2 Part 2: Transfer-Learning Comparison

In Part 2, head-only transfer achieved 0.68 test accuracy and test loss 1.275. Fine-tuning the final block achieved 0.52 accuracy and test loss 1.662. The head-only variant also achieved macro precision 0.755, recall 0.680, and F1 0.669. These outcomes indicate that CNN A's features were somewhat useful for the second task, but did not result in high classification performance. Additionally, fine-tuning deeper layers reduced rather than improved test performance.

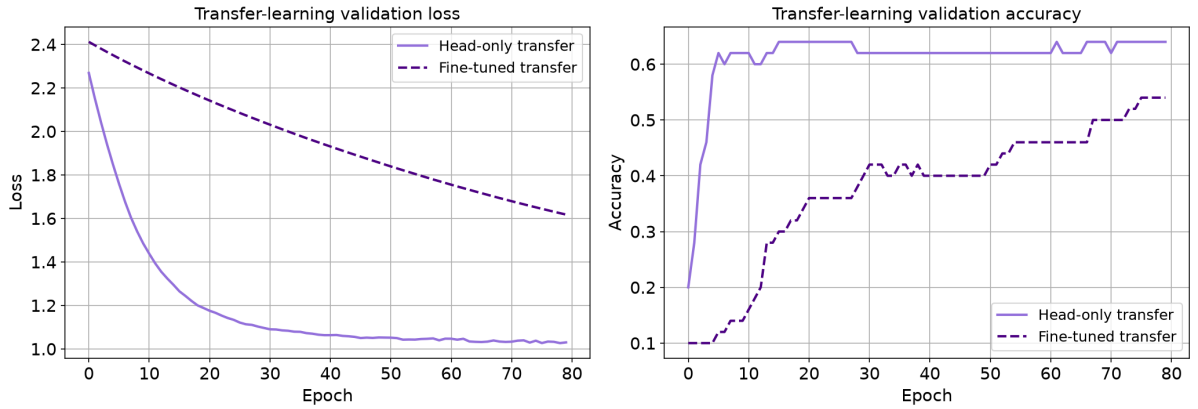



Figure 3: Validation accuracy and loss for the Part 2 transfer-learning variants.

## 4 Training Efficiency

CNN A reached peak validation accuracy at epoch 49; the head-only CNN B reached its peak at epoch 16, while the fine-tuned variant peaked at epoch 76. Although epoch count is not a direct measure of efficiency, it is indicative that less training was required to Further, the transfer-learning variants (head-only and fine-tuned) reused a frozen feature extractor and updated only its new classification head which reduced the trainable parameters. To select CNN A a hyperparameter search was required which is expensive as it involved training multiple models per search. Since elapsed training time and energy were not measured, a quantitative comparison of total compute or efficiency cannot be made. Although the transfer-learning model training updates were more efficient, the overall benefit was limited, so its additional optimisation was not justified by the measured accuracy.

## 5 Resource Estimate for 525 Species

Part 2 contains 1,561 training images for ten species, or about 156 images per species. If the 525-species dataset had a similar number of images per species, it would contain approximately

$$525 \times 156 \approx 81{,}900 \text{ images},$$

This is 52.5 times the current training set. With 224×224 images and a batch size of 32, training would take approximately 2,560 steps per epoch, compared with 49 steps currently. Therefore, a 50-epoch run would process about 4.1 million images.

Increasing the classifier from 10 to 525 outputs adds parameters to the final layer, but the convolutional feature extractor remains the main computation for each image. A practical full-scale experiment could use a pretrained backbone, a streamed input pipeline, class-weighted loss, and validation-based early stopping. Training should begin with a pilot run to measure throughput and memory use before committing to the full schedule.

This is an estimate: actual training time would depend on the hardware, model, and convergence behaviour. It also excludes validation, hyperparameter tuning, and repeat runs. Further, this assumes the observed average of approximately 156 training images per species; the actual dataset distribution may differ.

## 6 Conclusion

In Part 1, the selected four-block CNN A provided the strongest result for its 10-class task, with 98% test accuracy. In Part 2, its features transferred to a separate set of species with moderate success: training only a new head achieved 68% accuracy, while fine-tuning the final block achieved 52%. The head-only model reached peak validation accuracy in fewer epochs than CNN A and was more parameter-efficient in optimisation, although no timing data were available to quantify overall runtime savings. For a 525-species task, the observed sample rate implies approximately 82,000 training images and a substantial increase in per-epoch workload.